

Metrics. Tokenizer: reconstruction PSNR/SSIM on held-out frames. Action model: NMI_{adj} between action codes and VLM-judged action groups, all groups, on the frozen action-group set (#16); same protocol for every model, random ≈ 0 . Dynamics: one-step next-frame prediction from real context on the held-out split (Push-T: NanoWM protocol, 3 autoregressive steps). – = not relevant or not measured. † = continuous CoMo action (full 128-d z): may leak the target, read as an upper bound.

Zelda (128 px)

Dynamics: 890 held-out windows, 3 context frames $\rightarrow$ next frame, decoded with context, 1 MaskGIT pass / 1 Euler step. Tokenizer recon: 220 held-out 7-frame windows. LPIPS marked * is at 4 Euler steps.

Tokenizer			Action model		Dynamics				
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓	Notes
–	–	–	–	–	copy last frame	27.43	0.681	0.098	baseline
FSQ temporal	32.42	0.933	LAM (v3)	0.110	MaskGIT	26.72	0.759	–	original Genie pipeline (#4, #14)
FSQ temporal	32.42	0.933	warp LAM (WTA)	0.471	MaskGIT + copy prior	27.80	0.801	–	first to beat copy-last (#9, #14)
FSQ temporal	32.42	0.933	CoMo-ViT, 16 codes	0.561	MaskGIT	28.02	0.817	0.096	CoMo on MAE ViT-L features (#20)
FSQ temporal	32.42	0.933	CoMo-ViT, 16 codes	0.561	flow matching	28.34	0.816	0.099*	FM > MaskGIT twin (#19)
FSQ per-frame	32.12	0.929	CoMo-Tok, 16 codes	–	MaskGIT	27.39	0.789	0.109	CoMo on our tokenizer features (#21)
FSQ per-frame	32.12	0.929	CoMo-Tok, full z	–	MaskGIT	29.06 [†]	0.865	0.085	(#21)
FSQ temporal	32.42	0.933	CoMo-ViT, full z	0.561	flow matching	29.81 [†]	0.884	0.086	(#19)
continuous KL (16 ch)	44.46	0.995	CoMo-ViT, full z	0.561	flow matching	33.78 [†]	0.937	0.036	+4 dB from continuous tokens (#24)
continuous KL (16 ch)	44.46	0.995	CoMo-ViT, full z	0.561	FlowC2S (copy-last flow)	34.09[†]	0.940	0.034	fine-tuned from row above (#25)
continuous KL (16 ch)	44.46	0.995	CoMo-ViT, full z	0.561	Bi-flow	33.72 [†]	0.933	0.038	from scratch, 30k (#25)
–	–	–	CoMo-ViT, full z	0.561	CoMo decoder	33.41 [†]	0.931	0.038	CoMo decoder as dynamics (#20)
–	–	–	CoMo-ViT, 16 codes	0.561	CoMo decoder	30.15	0.844	0.050	(#20)

Merged training stages (STA-65) — in progress

NMI_{adj} for CoMo-ViT is the 50k encoder clustered into 16 codes (10 k-means seeds: 0.550 ± 0.010); camera-shift oracle 0.485. CoMo-Tok was only scored on the older Zelda judge set (moves NMI_{adj} 0.60 vs 0.74 for CoMo-ViT, #18).

Push-T (128 px, ground-truth actions)

NanoWM protocol: 21 val episodes, 256 clips, 1 real frame → 3 predicted autoregressively. NanoWM predictions re-scored with our scorer (validated against theirs).

Tokenizer			Action model		Dynamics				Notes
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓	
–	–	–	–	–	copy last frame	26.59	0.950	0.056	baseline
–	–	–	ground truth	–	NanoWM-B/2 (100k)	35.10	0.985	0.009	external reference
FSQ (4096 codes)	42.84	0.996	ground truth	–	MaskGIT, 30k	34.54	0.983	0.018	FID 26.0 vs 9.0 (#8)

Sonic (64 px)

Held-out split (10 blocks), 500 windows, 3 context frames → next frame, 10 MaskGIT passes. Tokenizer recon on the same windows.

Tokenizer			Action model		Dynamics				Notes
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓	
–	–	–	–	–	copy last frame	23.98	0.629	–	baseline; no model beats it
FSQ (author)	17.23	–	LAM (author)	–	MaskGIT (author)	11.72	0.314	–	upstream’s released weights
FSQ temporal	24.83	0.83	LAM (v3, 8 codes)	0.080	MaskGIT, 20k	18.57	0.583	–	(#2, #3)
FSQ temporal	24.83	0.83	none (learned null)	–	MaskGIT, 20k	18.36	0.572	–	actions add almost nothing
									(#3)

Pong (64 px)

Held-out split (10 blocks), 1340 windows, 3 context frames → next frame, 10 MaskGIT passes. Final checkpoint.

Tokenizer			Action model		Dynamics				Notes
	PSNR↑	SSIM↑		NMI _{adj} ↑		PSNR↑	SSIM↑	LPIPS↓	
–	–	–	–	–	copy last frame	30.06	0.953	–	baseline
FSQ temporal	45.70	0.997	LAM (16 codes)	0.416	MaskGIT, 39k	42.39	0.993	–	(#5)
FSQ temporal	45.70	0.997	random code	–	MaskGIT, 39k	35.95	0.978	–	same model, random action